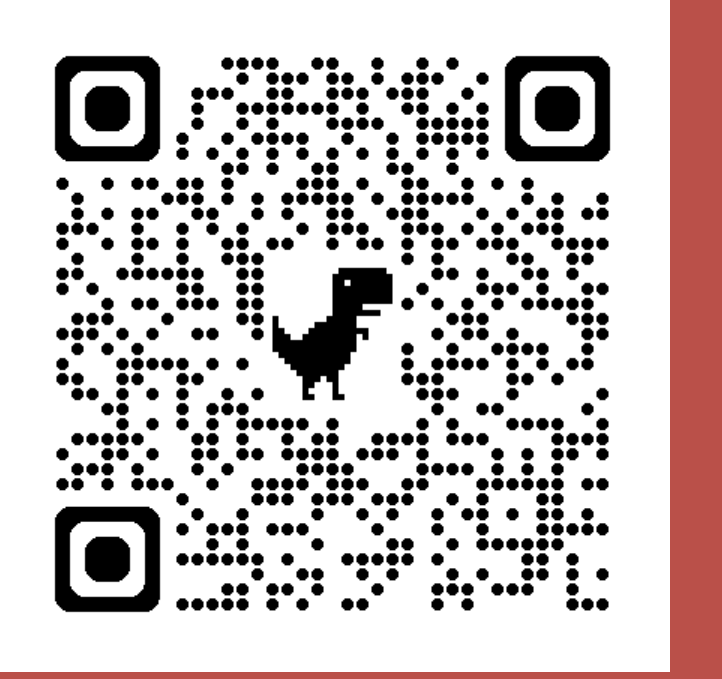


# Exploiting Similarity for Computation and Communication-Efficient Decentralized Optimization

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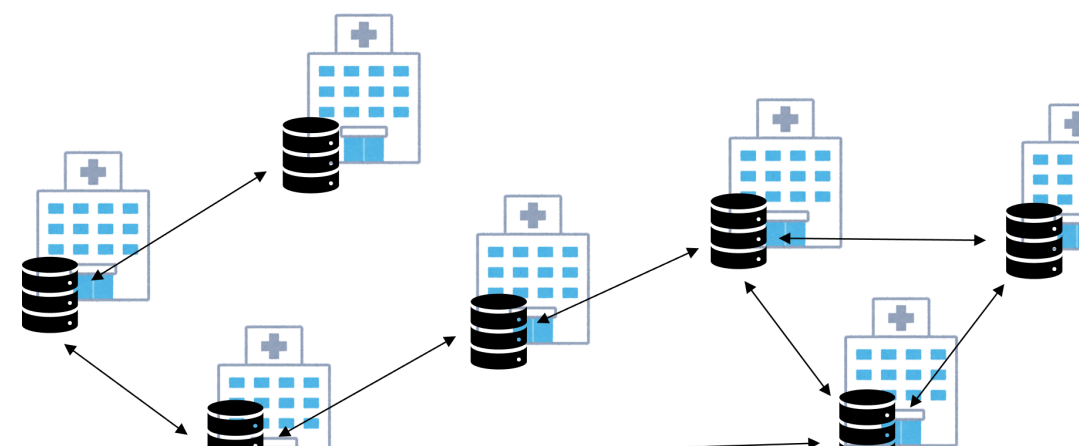
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## Background

### Decentralized learning

Let the number of nodes be  $n$  and the loss function of node  $i$  be  $f_i(x)$ , decentralized learning is formulated as follows:

$$\min_{x \in \mathbb{R}^d} \left[ f(x) := \frac{1}{n} \sum_{i=1}^n f_i(x) \right] \quad (1)$$


### Decentralized GD

$$x_i^{(t+1)} = \sum_{j=1}^n W_{ij} \left( x_j^{(t)} - \eta \nabla f_j(x_j^{(t)}) \right). \quad (2)$$

#### Assumption 1 (Strong Convexity)

$f_i$  is  $\mu$ -strongly convex.

#### Assumption 2 ( $L$ -smoothness)

$$\|\nabla f_i(x) - \nabla f_i(y)\| \leq L \|x - y\| \quad (\forall x, y \in \mathbb{R}^d) \quad (3)$$

#### Definition 1 (Similarity)

$$\frac{1}{n} \sum_{i=1}^n \|\nabla h_i(x) - \nabla h_j(y)\|^2 \leq \delta^2 \|x - y\|^2, \quad (4)$$

for any  $x, y \in \mathbb{R}^d$  where  $h_i(x) := f_i(x) - f(x)$ .

We have  $\delta \leq L$ . Especially when nodes have similar datasets,  $\delta \ll L$  holds.

#### Assumption 3 (Spectral Gap)

There exists  $p \in (0, 1]$  that satisfies

$$\sum_{i=1}^n \left\| \sum_{j=1}^n W_{ij} x_j - \bar{x} \right\|^2 \leq (1-p) \sum_{i=1}^n \|x_i - \bar{x}\|^2, \quad (5)$$

for any  $x_1, x_2, \dots, x_n \in \mathbb{R}^d$  where  $\bar{x} := \frac{1}{n} \sum_{i=1}^n x_i$ .

► **Contribution:** Summary of communication and computational complexities when  $\mu \leq \delta_{\max}$  and  $\mu \leq \delta$

| Algorithm                        | # Communication                                                                                                                                                              | # Computation                                                                                                                            |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Gradient Tracking                | $\mathcal{O}\left(\frac{L}{\mu(1-\rho)^2} \log\left(\frac{1}{\epsilon}\right)\right)$                                                                                        | $\mathcal{O}\left(\frac{L}{\mu(1-\rho)^2} \log\left(\frac{1}{\epsilon}\right)\right)$                                                    |
| Exact-PDO (SONATA)               | $\mathcal{O}\left(\frac{\delta}{\mu(1-\rho)} \log\left(\frac{L}{\delta}\right) \log\left(\frac{1}{\epsilon}\right)\right)$                                                   | n.a.                                                                                                                                     |
| Inexact-PDO [new]                | $\mathcal{O}\left(\frac{\delta}{\mu(1-\rho)} \log\left(\frac{L}{\delta}\right) \log\left(\frac{1}{\epsilon}\right)\right)$                                                   | $\mathcal{O}\left(\frac{\sqrt{\delta L}}{\mu} \log\left(\frac{1}{\epsilon}\right) \log\log\left(\frac{1}{\epsilon}\right)\right)$        |
| Stabilized-PDO [new]             | $\mathcal{O}\left(\frac{\delta}{\mu(1-\rho)} \log\left(\frac{L}{\delta}\right) \log\left(\frac{1}{\epsilon}\right)\right)$                                                   | $\mathcal{O}\left(\frac{\sqrt{\delta L}}{\mu} \log\left(\frac{1}{\epsilon}\right)\right)$                                                |
| Accelerated SONATA               | $\mathcal{O}\left(\sqrt{\frac{\delta_{\max}}{\mu(1-\rho)}} \log\left(\frac{L}{\delta}\right) \log\left(\frac{\delta}{\mu}\right) \log\left(\frac{1}{\epsilon}\right)\right)$ | n.a.                                                                                                                                     |
| Inexact Accelerated SONATA       | $\mathcal{O}\left(\sqrt{\frac{\delta_{\max}}{\mu(1-\rho)}} \log\left(\frac{L}{\delta}\right) \log\left(\frac{\delta}{\mu}\right) \log\left(\frac{1}{\epsilon}\right)\right)$ | $\mathcal{O}\left(\frac{\sqrt{\delta_{\max} L}}{\mu} \log\left(\frac{\delta}{\mu}\right) (\log\left(\frac{1}{\epsilon}\right))^2\right)$ |
| Accelerated Stabilized-PDO [new] | $\mathcal{O}\left(\sqrt{\frac{\delta}{\mu(1-\rho)}} \log\left(\frac{L}{\mu}\right) \log\left(\frac{1}{\epsilon}\right)\right)$                                               | $\mathcal{O}\left(\sqrt{\frac{L}{\mu}} \log\left(\frac{1}{\epsilon}\right)\right)$                                                       |

## Proposed Method

### Proximal Decentralized Optimization (PDO)

**Algorithm 1** Proximal Decentralized Optimization Method (PDO)

```

1: Input: The number of rounds  $R$ , and hyperparameters  $M$  and  $\lambda$ .
2: for  $r = 0, 1, \dots, R-1$  do
3:   for  $i = 1, \dots, n$  in parallel do
4:      $x_i^{(r+\frac{1}{2})} \approx \arg \min_x F_{i,r}(x)$  where  $F_{i,r}(x) := f_i(x) + \langle h_i^{(r)}, x \rangle + \frac{\lambda}{2} \|x - x_i^{(r)}\|^2$ .
5:      $x_i^{(r+1)} = \text{MULTIGOSSIP}\left(x_i^{(r+\frac{1}{2})}, W, M, i\right)$ .
6:      $h_i^{(r+1)} = \text{MULTIGOSSIP}\left(h_i^{(r)} + \nabla f_i(x_i^{(r+1)}), W, M, i\right) - \nabla f_i(x_i^{(r+1)})$ .
7:   end for
8: end for

```

Node  $i$  would like to minimize the following objective function:

$$f(x) = \underbrace{f_i(x)}_{\text{It's easy to minimize}} + \underbrace{f(x) - f_i(x)}_{\text{It's hard to minimize}}$$

$$\approx f_i(x) + \left\langle \nabla f(x^{(r)}) - \nabla f_i(x^{(r)}), x - x^{(r)} \right\rangle + \frac{\lambda}{2} \|x - x^{(r)}\|^2.$$

#### Theorem (PDO)

Suppose that Assumptions 1, 2, and 3 hold, and  $h_i^{(0)} = \nabla h_i(x_i^{(0)})$ ,  $x_i^{(0)} = \bar{x}^{(0)}$ , and

$$\sum_{i=1}^n \left\| \nabla F_{i,r}(x_i^{(r+\frac{1}{2})}) \right\|^2 \leq \frac{\delta(4\delta + \mu)}{4(r+1)(r+2)} \sum_{i=1}^n \left\| x_i^{(r+\frac{1}{2})} - x_i^{(r)} \right\|^2.$$

There exists  $\lambda$  and  $M$  such that  $\frac{1}{WR} \sum_{r=0}^{R-1} w^{(r)} (f(\bar{x}^{(r+1)}) - f(x^*)) \leq \epsilon$  is satisfied after

$$\mathcal{O}\left(\frac{\mu + \delta}{\mu(1-\rho)} \log\left(\frac{L}{\delta}\right) \log\left(\frac{\mu \|\bar{x}^{(0)} - x^*\|^2}{\epsilon}\right)\right)$$

communication where  $\bar{x}^{(r)} := \frac{1}{n} \sum_{i=1}^n x_i^{(r)}$ .

### Stabilized PDO (SPDO)

SPDO can achieve the same communication complexity as PDO if we satisfy:

$$\sum_{i=1}^n \left\| \nabla F_{i,r}(x_i^{(r+\frac{1}{2})}) \right\|^2 \leq 40\delta^2 \sum_{i=1}^n \left\| x_i^{(r+\frac{1}{2})} - x_i^{(r)} \right\|^2.$$

**Algorithm 3** Stabilized Proximal Decentralized Optimization Method (SPDO)

```

1: Input: The number of rounds  $R$ , and hyperparameters  $M$  and  $\lambda$ .
2: for  $r = 0, 1, \dots, R-1$  do
3:   for  $i = 1, \dots, n$  in parallel do
4:      $x_i^{(r+1)} \approx \arg \min_{x \in \mathbb{R}^d} F_{i,r}(x)$  where  $F_{i,r}(x) := f_i(x) + \langle h_i^{(r)}, x \rangle + \frac{\lambda}{2} \|x - v_i^{(r)}\|^2$ .
5:      $v_i^{(r+\frac{1}{2})} = \arg \min_{v \in \mathbb{R}^d} \left\{ \langle \nabla f_i(x_i^{(r+1)}), h_i^{(r)} \rangle + \frac{\mu}{2} \|v - x_i^{(r+1)}\|^2 + \frac{\lambda}{2} \|v - v_i^{(r)}\|^2 \right\}^a$ .
6:      $v_i^{(r+1)} = \text{MULTIGOSSIP}\left(v_i^{(r+\frac{1}{2})}, W, M, i\right)$ .
7:      $h_i^{(r+1)} = \text{MULTIGOSSIP}\left(h_i^{(r)} + \nabla f_i(v_i^{(r+1)}), W, M, i\right) - \nabla f_i(v_i^{(r+1)})$ .
8:   end for
9: end for

```

<sup>a</sup>This update rule can be rewritten as  $v_i^{(r+\frac{1}{2})} = \frac{1}{\mu+\lambda} (\mu x_i^{(r+1)} + \lambda v_i^{(r)} - \nabla f_i(x_i^{(r+1)}) - h_i^{(r)})$ .

### Accelerated SPDO

**Algorithm 4** Accelerated Stabilized Proximal Decentralized Optimization Method (Accelerated-SPDO)

```

1: Input: The number of rounds  $R$ , and hyperparameters  $\gamma$ ,  $M$ , and  $\lambda$ .
2:  $A_0 = 0$  and  $B_0 = 1$ .
3: for  $r = 0, 1, \dots, R-1$  do
4:   Find  $a_{r+1} > 0$  that satisfies  $\lambda = \frac{(A_r + a_{r+1})B_r}{a_{r+1}^2}$ .
5:    $A_{r+1} = A_r + a_{r+1}$ .
6:    $B_{r+1} = B_r + \mu a_{r+1}$ .
7:   for  $i = 1, \dots, n$  in parallel do
8:      $y_i^{(r)} = \frac{A_r x_i^{(r)} + a_{r+1} v_i^{(r)}}{A_{r+1}}$ .
9:     if  $r \geq 1$  then
10:       $h_i^{(r)} = \text{FASTGOSSIP}\left(h_i^{(r-1)} + \nabla f_i(y_i^{(r)}), W, M, \gamma, i\right) - \nabla f_i(y_i^{(r)})$ .
11:    end if
12:     $x_i^{(r+\frac{1}{2})} \approx \arg \min_{x \in \mathbb{R}^d} F_{i,r}(x)$  where  $F_{i,r}(x) := f_i(x) + \langle h_i^{(r)}, x \rangle + \frac{\lambda}{2} \|x - y_i^{(r)}\|^2$ .
13:     $v_i^{(r+\frac{1}{2})} = \arg \min_{v \in \mathbb{R}^d} \left\{ a_{r+1} \left( \langle \nabla f_i(x_i^{(r+\frac{1}{2})}), h_i^{(r)} \rangle + \frac{\mu}{2} \|v - x_i^{(r+\frac{1}{2})}\|^2 \right) + \frac{B_r}{2} \|v - v_i^{(r)}\|^2 \right\}^a$ .
14:     $x_i^{(r+1)} = \text{FASTGOSSIP}\left(x_i^{(r+\frac{1}{2})}, W, M, \gamma, i\right)$ .
15:     $v_i^{(r+1)} = \text{FASTGOSSIP}\left(v_i^{(r+\frac{1}{2})}, W, M, \gamma, i\right)$ .
16:  end for
17: end for

```

<sup>a</sup>This update rule can be rewritten as  $v_i^{(r+\frac{1}{2})} = \frac{1}{B_{r+1}} (B_r v_i^{(r)} + \mu a_{r+1} x_i^{(r+\frac{1}{2})} - a_{r+1} (\nabla f_i(x_i^{(r+\frac{1}{2})}) + h_i^{(r)}))$ .

**Algorithm 2** Multiple Gossip Averaging

```

1: function MULTIGOSSIP( $\{a_i\}_{i=1}^n, W, M, i$ )
2:    $a_i^{(0)} = a_i$  for all  $i$ .
3:   for  $m = 0, \dots, M-1$  do
4:      $a_i^{(m+1)} = \sum_{j=1}^n W_{ij} a_j^{(r)}$ .
5:   end for
6:   return  $a_i^{(M)}$ 
7: end function

```

**Algorithm 5** Fast Gossip Averaging

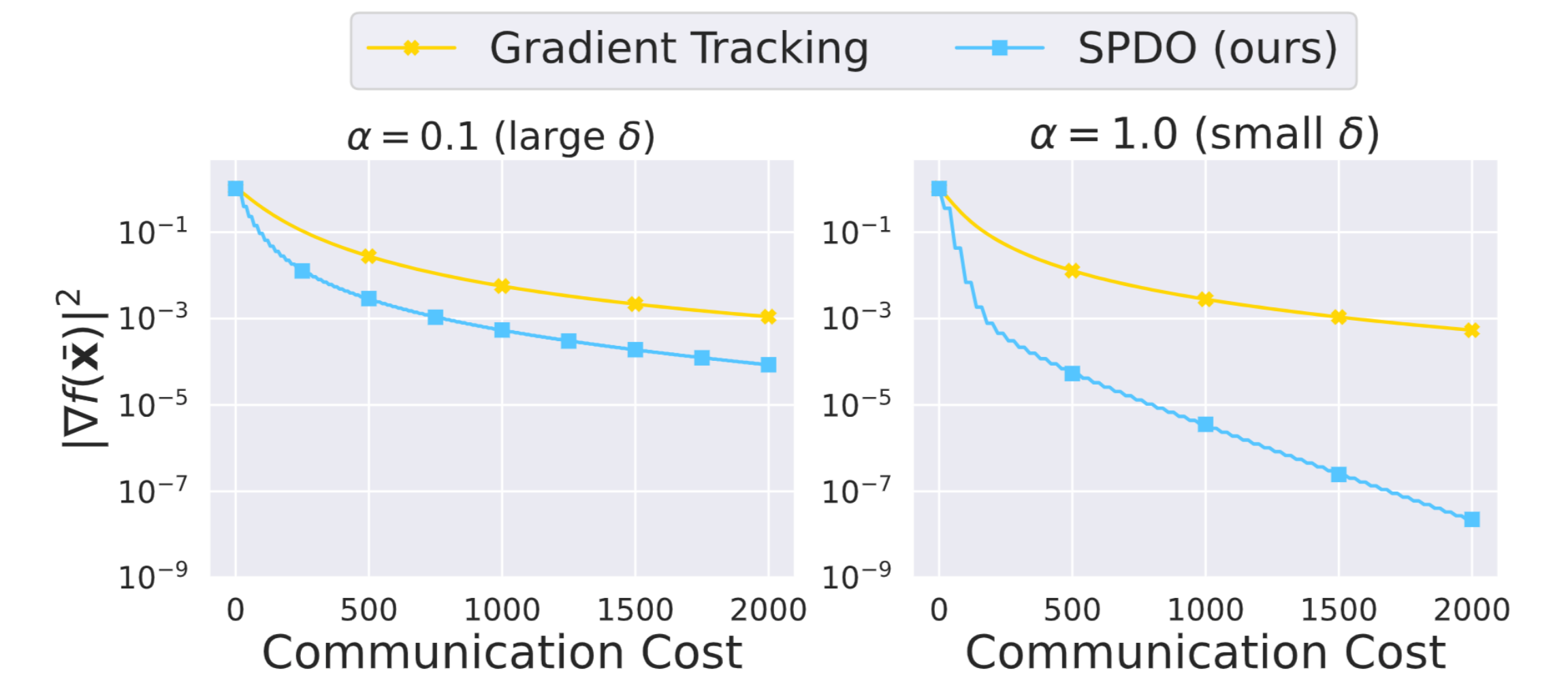
```

1: function FASTGOSSIP( $\{a_i\}_{i=1}^n, W, M, \gamma, i$ )
2:    $a_i^{(0)} = a_i$  and  $a_i^{(-1)} = a_i$  for all  $i$ .
3:   for  $m = 0, \dots, M-1$  do
4:      $a_i^{(m+1)} = (1 + \gamma) \sum_{j=1}^n W_{ij} a_j^{(m)} - \gamma a_i^{(m-1)}$ .
5:   end for
6:   return  $a_i^{(M)}$ .
7: end function

```

## Experiments

► SPDO can exploit the similarity and reduce the communication costs.



► Acc. SPDO can achieve the best communication and computation complexities.

